

## The Chain Rule — From Calc I to Calc III

In Calc I, the chain rule handled **compositions** of single-variable functions:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

**Example:** If  $y = \sin(x^2)$ , let  $u = x^2$ . Then  $y = \sin u$ , so:

$$\frac{dy}{dx} = \cos(u) \cdot 2x = 2x \cos(x^2)$$

**Now in Calc III:** What if  $z$  depends on  $x$  and  $y$ , and  $x$  and  $y$  each depend on other variables?

There are **multiple paths** from the independent variable to  $z$  — we need to account for **all of them**.

Calc I chain rule: one path, one product. Calc III chain rule: **multiply along each branch, add across branches**.

### Case 1: One Independent Variable

Suppose  $z = f(x, y)$  where  $x = g(t)$  and  $y = h(t)$ . Then  $z$  is ultimately a function of  $t$  alone.

#### Chain Rule (Case 1)

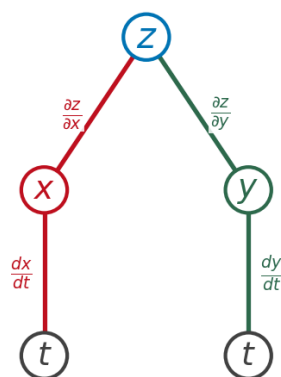
If  $z = f(x, y)$  is differentiable, and  $x = g(t)$ ,  $y = h(t)$  are differentiable functions of  $t$ , then:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

Notice the pattern:

- One term for **each intermediate variable** ( $x$  and  $y$ )
- Each term is a product: (partial of  $z$ )  $\times$  (derivative of that variable)
- The terms are **added** together

A **variable tree** makes the pattern visual. Draw the dependencies, then read off the formula:



#### How to read the tree:

1. Each **branch** from top to bottom is a path from  $z$  to  $t$
2. **Multiply** the derivatives along each branch
3. **Add** the products from all branches

**Left branch:**  $\frac{\partial z}{\partial x} \cdot \frac{dx}{dt}$

**Right branch:**  $\frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$

**Total:**  $\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$

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Draw the tree **first**, then read off the formula. **Multiply along each path, add across paths**.

## Example 1: One Independent Variable

### Example 1

Let  $z = x^2 + y^2 + xy$ , where  $x = \sin t$  and  $y = e^t$ .

Find  $\frac{dz}{dt}$  at  $t = \frac{\pi}{4}$ .

*Draw the variable tree, write the chain rule formula, compute each piece, then evaluate at  $t = \pi/4$ .*

## Your Turn: One Variable Chain Rule

### Example 2

Let  $z = e^x \sin y$ , where  $x = t^2$  and  $y = 3t$ .

Find  $\frac{dz}{dt}$ .

*Draw the variable tree, compute the partials and derivatives, then combine.*

## Case 2: Two Independent Variables

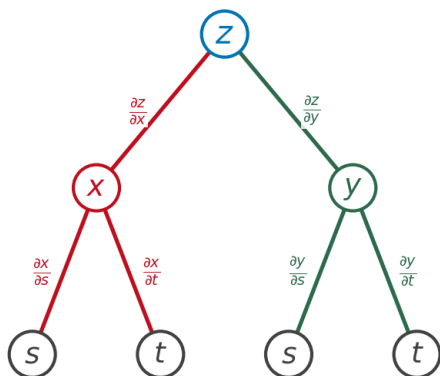
Now suppose  $z = f(x, y)$  where  $x = g(s, t)$  and  $y = h(s, t)$ . Then  $z$  depends on **two** independent variables  $s$  and  $t$ .

### Chain Rule (Case 2)

If  $z = f(x, y)$ ,  $x = g(s, t)$ ,  $y = h(s, t)$ , all differentiable, then:

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$



The tree now has **four branches** — two paths to  $s$  and two paths to  $t$ .

For  $\partial z / \partial s$ : follow only the branches that end at  $s$ , multiply along each, and add.

Same idea — but now each intermediate variable ( $x, y$ ) depends on **both**  $s$  and  $t$ .

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With two independent variables, we get **two formulas** — one for  $\partial z / \partial s$  and one for  $\partial z / \partial t$ . Each follows the same tree pattern.

### Example 3: Two Independent Variables

#### Example 3

Let  $z = x^2 y^3$ , where  $x = s \cos t$  and  $y = s \sin t$ .

Find  $\frac{\partial z}{\partial s}$  when  $(s, t) = (2\sqrt{2}, \pi/4)$ .

Draw the variable tree for  $\partial z / \partial s$ . Compute partials, then evaluate at the given point.

### The General Chain Rule

The pattern extends to **any** number of intermediate and independent variables.

### General Chain Rule

If  $u = f(x_1, x_2, \dots, x_n)$  where each  $x_i = g_i(t_1, t_2, \dots, t_m)$ , then:

$$\frac{\partial u}{\partial t_j} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_j} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial t_j} + \dots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_j}$$

for each  $j = 1, 2, \dots, m$ .

**The recipe is always the same:**

1. Draw the variable tree
2. For each path from  $u$  down to  $t_j$ : multiply the derivatives along the path
3. Add all the path-products together

You don't need to memorize separate formulas for each case. Just **draw the tree** and read off the answer.

### Implicit Differentiation Revisited

In Calc I, we used implicit differentiation to find  $dy/dx$  when  $y$  was defined implicitly by an equation like  $x^2 + y^2 = 25$ .

In Calc III, we often have an equation like  $F(x, y, z) = 0$  that defines  $z$  implicitly as a function of  $x$  and  $y$ . How do we find  $\partial z / \partial x$ ?

**Method 1 — Direct differentiation:**

- Treat  $y$  as a constant and  $z$  as a function of  $x$
- Differentiate the equation with respect to  $x$
- Solve for  $\partial z / \partial x$

**Method 2 — The shortcut formula:**

If  $F(x, y, z) = 0$  defines  $z$  implicitly as a function of  $x$  and  $y$ , and  $F_z \neq 0$ :

### Implicit Differentiation Formulas

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

Method 2 is derived **from** the chain rule. It's a convenient shortcut — but you should understand why it works.

### Why Does $\partial z / \partial x = -F_x / F_z$ Work?

Start with  $F(x, y, z) = 0$ , where  $z = z(x, y)$  is implicitly defined.

Differentiate both sides with respect to  $x$  (holding  $y$  constant):

By the chain rule:

$$\frac{\partial F}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} = 0$$

Since  $x$  is independent:  $\frac{\partial x}{\partial x} = 1$ . Since  $y$  is held constant:  $\frac{\partial y}{\partial x} = 0$ .

$$F_x \cdot 1 + F_y \cdot 0 + F_z \cdot \frac{\partial z}{\partial x} = 0$$

Solve for  $\partial z / \partial x$ :

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \quad \square$$

The formula comes directly from the chain rule applied to  $F(x, y, z) = 0$ . Same idea gives  $\partial z / \partial y = -F_y / F_z$ .

### Example 4: Implicit Differentiation

#### Example 4

The equation  $x^2 \sin(yz) + 3z = y^2$  defines  $z$  implicitly as a function of  $x$  and  $y$ .

Find  $\frac{\partial z}{\partial x}$  using both methods.

Method 1: Differentiate directly w.r.t.  $x$  ( $y$  constant,  $z = z(x, y)$ ). Method 2: Set  $F = 0$ , use  $\partial z / \partial x = -F_x / F_z$ .

### Your Turn: Implicit Differentiation

#### Example 5

The equation  $x^3 + y^3 + z^3 + 6xyz = 1$  defines  $z$  implicitly as a function of  $x$  and  $y$ .

Find  $\frac{\partial z}{\partial x}$  using the formula  $\partial z / \partial x = -F_x / F_z$ .

Set  $F(x, y, z) = 0$ , compute  $F_x$  and  $F_z$ , then apply the formula.

### Homework

**Section 14.5:** 5, 7, 13, 15, 17, 19, 25, 35, 37, 39, 44, 49